CHARMINE YANG

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PROFILE

Cybersecurity and Digital Forensics student at Nanyang Polytechnic (GPA 3.83/4.00) with hands-on experience in threat intelligence visualization, AI-augmented phishing simulations, and CTF challenge development. Active leader in AI and Infosec communities, combining strong technical fundamentals in networking and Python with proven teamwork and communication skills.

SKILLS

Languages
Python/Python Flask • HTML5 • CSS3 • JavaScript • SQL • Bash

Platforms & Tools
Wireshark • Nmap • Cisco Packet Tracer • Virtual Machines • MySQL Workbench • PowerBI Desktop

Core Competencies
Enterprise Networking • Web Applications • Threat Intelligence Visualization • Database Management

Soft Skills
Analytical Thinking • Problem Solving • Adaptability • Strong Willingness to Learn

PROJECTS

Threat Intelligence Visualization using PowerBI Desktop

- Architected a robust relational data model correlating 40,000+ firewall logs with external threat intel sources in PowerBI to support faster incident triage.
- Engineered interactive decomposition trees and forensic heatmaps to isolate high-risk protocol outliers (TCP/UDP), reducing alert fatigue and accelerating root-cause discovery.

AI-Augmented Phishing Framework | Singapore Cybersecurity Camp 2025

- Built an advanced phishing simulation tool to raise employee threat awareness by building an automated framework using OSINT scrapers and the Gemini API.
- Generated high-fidelity, psychologically targeted test emails to identify human-centric vulnerabilities within an organization.

NUS Illustratum Physics Competition 2024

- Researched the two-balloon airflow phenomenon and presented findings to NUS lecturers.
- Translated difficult physics principles into simple explanations for lecturers and peers, highlighting a drive for accessible knowledge sharing.

Engineering with Business Internship @ NYP 2023

- Built an automated traffic light simulation to replace manual logic with a streamlined, sensor-based solution.
- Programmed an Arduino using Python to integrate IR and ultrasonic sensors to demonstrate real-time automation and distance-based logic control; prototyped circuits in Tinkercad.
- Collaborated in a team to troubleshoot hardware and code issues, explored sensor-based input/output systems, and demonstrated a commitment to robust logic structuring.

EDUCATION

Cybersecurity and Digital Forensics (Dip) | Nanyang Polytechnic | Apr 2025-Present

- cGPA: 3.83/4.00; Recipient of the Director's List (Y1S1, Y1S2)
- Vice-President in NYP Infosec SIG
- Vice-President in NYP AI SIG
- Class Representative SF2501 AY26/27
- Volunteering under SG Cyber Safe Youth Network (CSA)
- Attending Sentinel Programme organized by the Digital and Intelligence Service (DIS): Selected to undergo advanced technical training in national-level defense competencies - Attained SHIELD Award
- Selected for CSIT x Sentinel Work Experience Programme, where I pentested CTF challenges for Cyberthon to find insecure workarounds that could be improved.
- Sentinel Challenge @ Brainhack 2026 - Champion for Poly/ITE category (Pulsar)

Bartley Secondary School | 2021-2024

- Internship experience at NYP Engineering with Business 2023 – Explored Arduino to bring ideas to life using Python codes across 4 weeks and applied them to real-world problems
- Bartley Scholar – Awarded Certification of Merit for ICAS Science Assessments (2021, 2022)
- Top in Cohort – Science (Sec 1) and Additional Mathematics (Sec 3)
- NUS Illustratum Physics Competition – Certificate of Merit
- MOE Edusave Scholarship – Recipient for four consecutive years (2021–2024)
- MOE EAGLES Award – Recognised for leadership and service (2024)
- MOE Character Award – Honored for exemplary conduct and values (2024)
- Student Councillor – Led and supported school-wide events and external volunteer initiatives
- Class Chairperson - Facilitated communication

LEADERSHIP & SERVICE

Executive Leadership

Vice-President | NYP Infosec SIG June 2025 - Present

- Engineered diverse CTF (Capture The Flag) challenges across multiple categories, leveraging the process to conduct deep-dive research into various security vulnerabilities and exploitation techniques.
- Represented Nanyang Polytechnic as a student ambassador at major industry events, including GovWare (Singapore International Cybersecurity Week) and SMU SkillsFuture Showcase, where I facilitated booth activities and engaged with the wider cybersecurity community to promote digital safety awareness.

Vice-President | NYP AI SIG May 2025 - Present

- Spearheaded community-building initiatives to strengthen subcommittee bonds and increase member retention, creating a cohesive environment for technical collaboration.
- Co-managed the execution of AI workshops for the student body, focusing on spreading AI literacy to peers and ensuring high-quality engagement through proactive logistical support.

Subcommittee Member | NYP School of Information Technology Club June 2025 - April 2026

- Actively engaged with the SIT student body to foster cross-disciplinary networking and peer-to-peer knowledge sharing within the School of Information Technology.
- Served as Program In-Charge (IC) for SITxSCS Career Mentoring Session, where I architected the complete program flow and curated professional emcee scripts to ensure a high-quality networking experience for students and industry mentors.

Community Volunteer | SG Cyber Safe Youth Network (CSA)

- Guided senior citizens through digital safety quizzes and interactive outreach activities monthly.
- Simplified security concepts using analogies and bilingual communication to bridge the digital literacy gap in the community.